Year 9 Mathematics

Unit 2: Area and Volume

2.4 Area of a Circle

Area of a Circle

STARTER · 5 MIN

Date: _____ **Title: Area of a Circle**

LEARNING OBJECTIVES

- Calculate the area of a circle
- Give areas in terms of π, and find areas of part circles
- Find shaded areas and the radius from the area

STARTER *Answer in your book. No calculators.*

1. Work out 6^2

2. Work out $\sqrt{49}$

3. The diameter is 18 cm. What is the radius?

4. Round 3.14159×4 to 1 d.p.

Starter — Answers

STARTER

Tick or correct in green pen

1. **36**

2. **7**

3. **9 cm**

4. **12.6**

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can calculate the area of a circle	☐
SILVER	I can give areas in terms of π, and find areas of part circles	☐
GOLD	I can find shaded areas and the radius from the area	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Area of a Circle

INTRODUCTION · 3 MIN

The area of a circle is π times the radius squared.

Formula

Square the radius, then times by π.

$A = \pi r^2$

Diameter given

Halve it first.

d = 10 cm: r = 5 cm

In terms of π

Leave π in the answer.

r = 8 cm: $A = 64\pi$ cm²

Part circles

Semicircle: half. Quarter circle: a quarter.

semicircle: $A = \frac{1}{2}\pi r^2$

Examples 1 and 2 — Area from the Radius or Diameter

EXAMPLES · 5 MIN

BRONZE

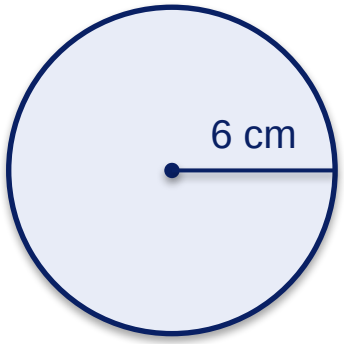
Watch and copy into your book

Hint $A = \pi r^2$ $r = d \div 2$

Example 1

Find the area.

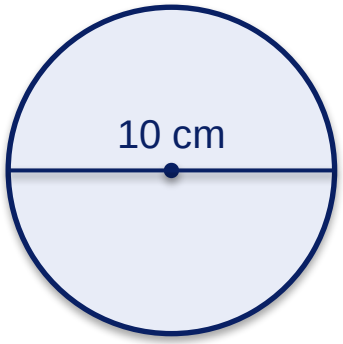
Give your answer to 1 d.p.



Example 2

Find the area.

Give your answer to 1 d.p.



Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

$A = \pi r^2$

$A = \pi \times 6^2$

$= \pi \times 36 = 113.097\ldots$

$= 113.1\ \text{cm}^2$ (1 d.p.)

Tip: square first, then times by π.

Example 2

Halve the diameter: r = 5

$A = \pi \times 5^2$

$= 78.539\ldots$

$= 78.5\ \text{cm}^2$ (1 d.p.)

Tip: $\pi \times 10^2$ uses the wrong length.

Task 1

PRACTICE · 9 MIN

BRONZE

Find the area. Give your answers to 1 d.p.

Hint $A = \pi r^2$ $r = d \div 2$

1. radius 3 cm
2. radius 10 cm
3. diameter 8 cm
4. diameter 14 cm
5. radius 2.5 cm

★ diameter 1.2 m

Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1. **28.3 cm²**
2. **314.2 cm²**
3. **50.3 cm²**
4. **153.9 cm²**
5. **19.6 cm²**

★ **1.1 m²**

Examples 3 and 4 — In Terms of π and Part Circles

EXAMPLES · 5 MIN

SILVER

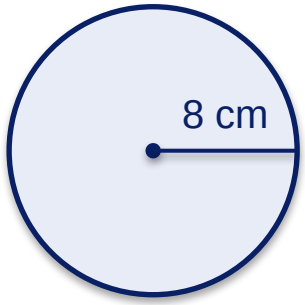
Watch and copy into your book

Hint $A = \pi r^2$ semicircle: $\frac{1}{2}\pi r^2$ quarter: $\frac{1}{4}\pi r^2$

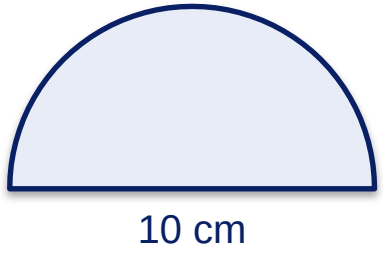
Example 3

Find the area in terms of π.

(a)

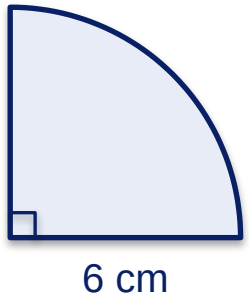


(b)



Example 4

Find the area of the quarter circle to 1 d.p.



Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

(a) circle (b) semicircle

(a) $\pi \times 8^2 = 64\pi \text{ cm}^2$

(b) $r = 5$: $\frac{1}{2} \times \pi \times 5^2$

$= 12.5\pi \text{ cm}^2$

Tip: a semicircle is half a circle.

Example 4

A quarter of a circle

$\frac{1}{4} \times \pi \times 6^2 = 9\pi$

$= 28.274\ldots$

$= 28.3 \text{ cm}^2$ (1 d.p.)

Tip: 36 ÷ 4 = 9

Task 2

PRACTICE · 9 MIN

SILVER

Questions 1–4: in terms of π. 5–6: to 1 d.p.

Hint $A = \pi r^2$ semicircle: $\frac{1}{2}\pi r^2$ quarter: $\frac{1}{4}\pi r^2$

1. Circle, radius 5 cm
2. Circle, diameter 12 cm
3. Semicircle, diameter 8 cm
4. Semicircle, radius 3 cm
5. Quarter circle, radius 4 cm

★ Quarter circle, radius 10 cm

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1. 25π cm²
2. 36π cm²
3. 8π cm²
4. 4.5π cm²
5. 12.6 cm²

★ 78.5 cm²

Examples 5 and 6 — Shaded Areas and Finding r

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

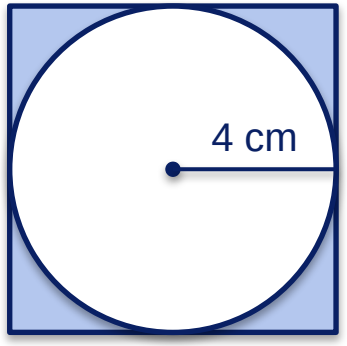
Hint shaded = square − circle $r^2 = A \div \pi$

Example 5

(3 marks)

Find the shaded area.

Give your answer to 1 d.p.



Example 6

(3 marks)

A circle has area 200 cm^2.

Find its diameter to the nearest cm.

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

Square − circle

square: $8 \times 8 = 64$

circle: $\pi \times 4^2 = 50.265\ldots$

$64 - 50.265\ldots = 13.7\ \text{cm}^2$

Tip: the side of the square is 2r.

Example 6

Work backwards: $r^2 = A \div \pi$

$r^2 = 200 \div \pi = 63.66\ldots$

$r = 7.978\ldots$

$d = 15.95\ldots = 16\ \text{cm}$

Tip: square root, then double.

Task 3

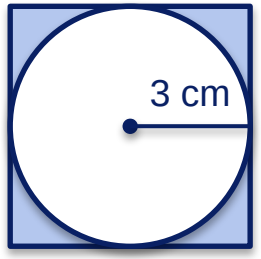
PRACTICE · 9 MIN

GOLD

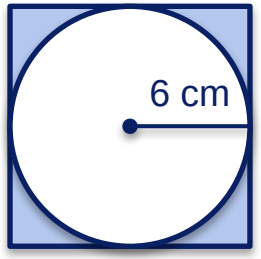
Questions 1–3: shaded area to 1 d.p. 4–6: work backwards.

Hint
shaded = square − circle
$r^2 = A \div \pi$

1 Shaded area



2 Shaded area



3 Shaded area



4 Find r (1 d.p.)

A circle has area 50 cm^2.

5 Find d

A circle has area 314 cm^2.
Give d to the nearest cm.

★ Find d

A plate has area 1000 cm^2.
Give d to the nearest cm.

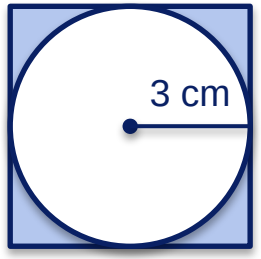
Task 3 — Answers

CORRECTIONS

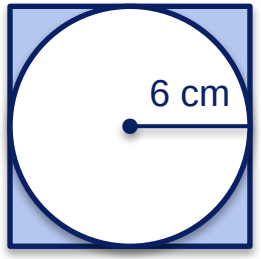
GOLD

Tick or correct in green pen

1 **7.7 cm²**



2 **30.9 cm²**



3 **5.4 cm²**



4 **r = 4.0 cm**

$r^2 = 50 \div \pi = 15.91\ldots$

$r = 3.98\ldots = 4.0$ cm

5 **d = 20 cm**

$r^2 = 314 \div \pi = 99.9\ldots$

$r = 9.99\ldots$, $d = 20$ cm

★ **d = 36 cm**

$r^2 = 1000 \div \pi = 318.3\ldots$

$r = 17.84\ldots$, $d = 36$ cm

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Find the area: radius 4 cm (1 d.p.)

SILVER Area of a semicircle, diameter 6 cm, in terms of π

GOLD A circle has area 100 cm^2. Find r (1 d.p.).

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE **50.3 cm²**

SILVER **4.5π cm²**

GOLD **5.6 cm**

Exam Practice — Area of a Circle

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: show all your working

Hint $A = \pi r^2$ $r = d \div 2$

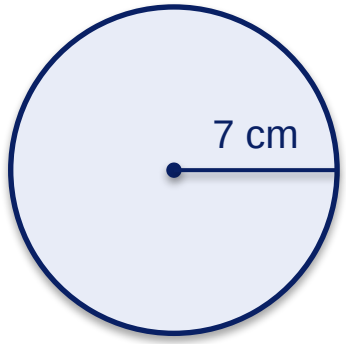
Question 1

(2 marks)

Find the area of this circle.

Give your answer to 1 d.p.

Diagram NOT accurately drawn



..... cm^2

Question 2

(2 marks)

A circular rug has a diameter of 3 m.

Find its area to 1 d.p.

..... m^2

Exam Practice — In Terms of π

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

Hint $A = \pi r^2$ semicircle: $\frac{1}{2}\pi r^2$ quarter: $\frac{1}{4}\pi r^2$

Question 3

(2 marks)

Find the areas in terms of π.

(a) a circle with radius 9 cm (1)

..... cm^2

(b) semicircle, diameter 14 cm (1)

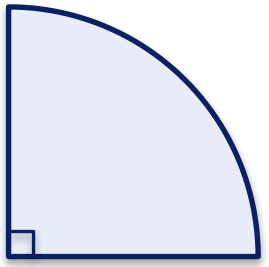
..... cm^2

Question 4

(2 marks)

Find the area of this quarter circle to 1 d.p.

Diagram NOT accurately drawn



..... cm^2

Exam Practice — Shaded Areas

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

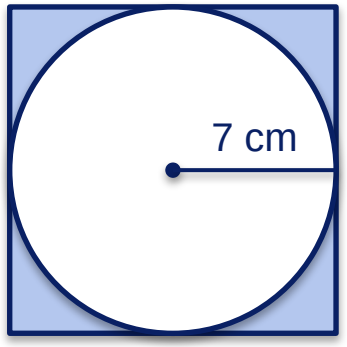
Hint shaded = square − circle $r^2 = A \div \pi$

Question 5

(3 marks)

Find the area of the shaded region.
Give your answer to 1 d.p.

Diagram NOT accurately drawn



..... cm^2

Question 6

(3 marks)

The area of a circular plate is 960 cm^2.
Calculate the diameter of the plate
to the nearest cm.

..... cm

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

$\pi \times 7^2$ — M1

$= 153.9\ cm^2$ — A1

Tip: square the radius first.

Question 2

$r = 1.5,\ \pi \times 1.5^2$ — M1

$= 7.1\ m^2$ — A1

Tip: halve the diameter.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

(a) 81π cm² B1

(b) 24.5π cm² B1

Tip: (b) r = 7, half of 49π

Question 4

$\frac{1}{4} \times \pi \times 12^2$ M1

= 113.1 cm² A1

Tip: 144 ÷ 4 = 36, so 36π

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

square = 14 × 14 = 196	M1
circle = π × 7^2 = 153.93…	M1
196 – 153.93… = 42.1 cm²	A1

Tip: the side of the square is 2 × 7

Question 6

960 ÷ π = 305.5…	M1
r = 17.48…	M1
d = 35 cm	A1

Tip: square root, then double.